GISHM Model

Figures 1 illustrate the software installation file (github repository doesn't contain installation file, for github users: clone repository and follow from figure 2), while Figure 2 shows the software icon displayed on the desktop after installation.

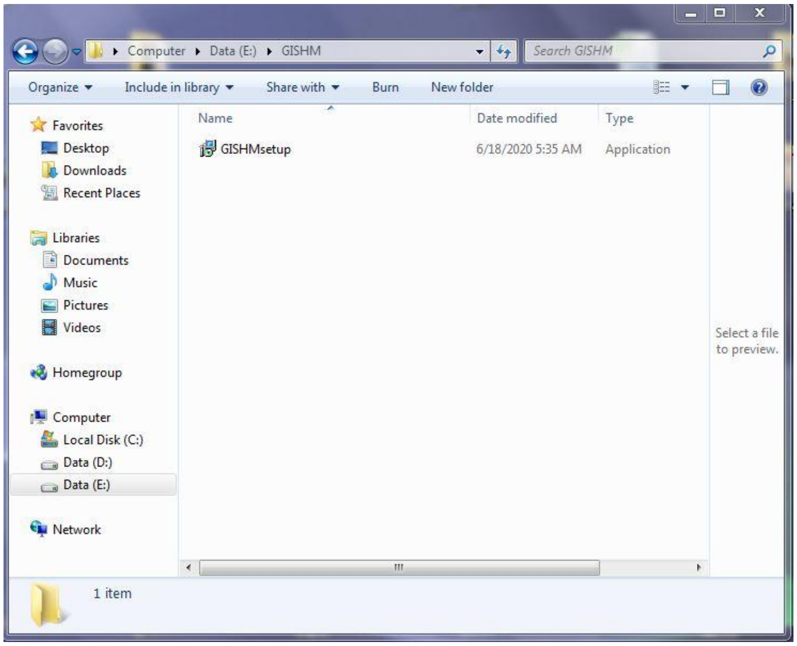


Fig1. Install GISHM(click on GISHMsetup)

Fig2. Start the application (click on GISHM icon)

2. Running the Model

2.1. First Model Tab (Input/Output)

The firs tab of the model contains the model input data and output directory settings (Figure 3). First, the **working directory** of the project should be specified. Subsequently, the required model inputs, including raster maps and time-series data, are assigned to the model.

To facilitate and accelerate the input configuration process, if all input files have the same names as those provided in the sample dataset distributed with the model, the software automatically loads (**autoloads**) all required input files from their corresponding locations once the project working directory is specified. Otherwise, users may manually assign each input file using custom filenames and file locations.

For automatic data loading, an **Inputs** folder must be created within the project **Working Directory**, following the same directory structure as the sample dataset. This folder should contain two subfolders named **maps** and **timeseries**, which store all required raster maps and time-series input files, respectively (Figure 4).

The **maps** folder must contain six raster maps (all using the same projected metric coordinate system) in ASCII raster format (*.asc*), with the following filenames (Figure 5):

- **Elevation** (Digital Elevation Model, elevation values in meters)
- **Landuse** (Land Use)
- **Soil** (Soil Type)
- **Thiessen_P** (Thiessen Precipitation Zones)
- **Thiessen_T** (Thiessen Temperature Zones)
- **Isochron** (Isochrone Map)

The **timeseries** folder must contain four time-series input files in *.tbl* format (Figure 6):

- **P** (Precipitation)
- **Q** (Discharge)
- **Tmax** (Maximum Air Temperature)
- **Tmin** (Minimum Air Temperature)

All raster maps must have identical spatial properties, including geographic extent, number of rows and columns, and coordinate reference system.

Raster datasets are assigned in the **Input Maps** section, while time-series files are specified in the **Input Timeseries** section.

Under the **Simulation Times** section, users should define the **Start Time Step**, **Final Time Step**, and the total number of **Time Steps**, expressed in days.

The **Output** section is used to specify the project output directory and the name of the output time-series file generated by the simulation in *.tbl* format. By default, the output file is named **results**.

When the **Input: Working Directory** is specified, the software automatically assigns the same location as the **Output: Output Directory** by default. Users may modify this location if they wish to save the model outputs to a different directory.

After model execution, the software automatically creates an **Outputs** folder within the selected output directory. All simulation outputs, including raster maps and the output time-series file (**results**), are stored in this folder.

It should be noted that the **Outputs** folder is automatically deleted before each new model run and recreated during execution, ensuring that only the outputs from the most recent simulation are retained.

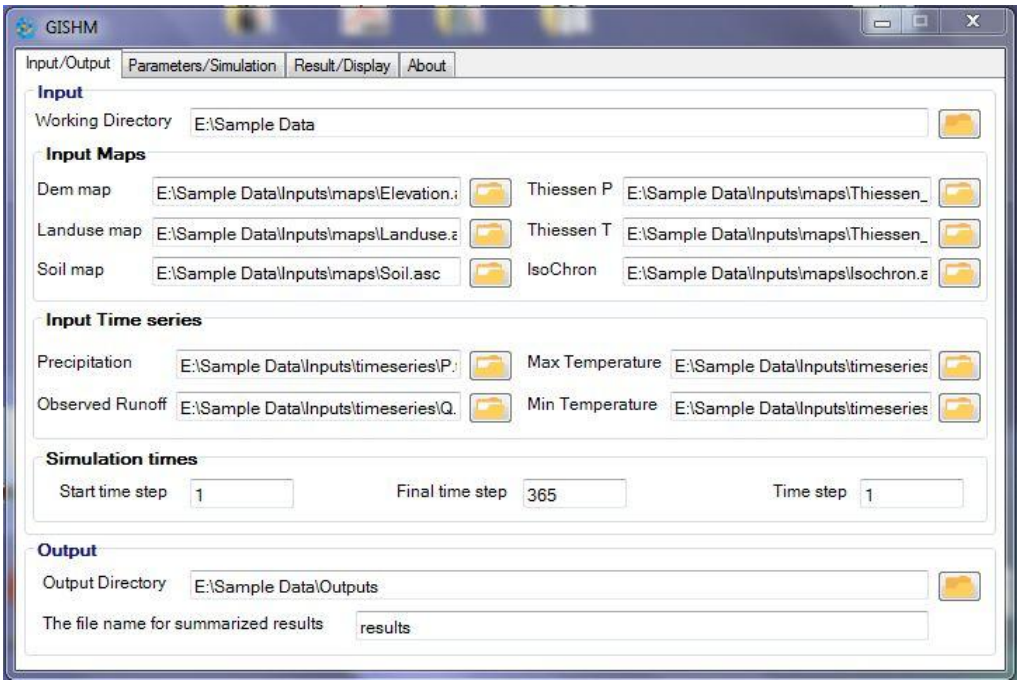


Figure 3. The first tab (Input/Output): Model inputs, outputs, and project working directory

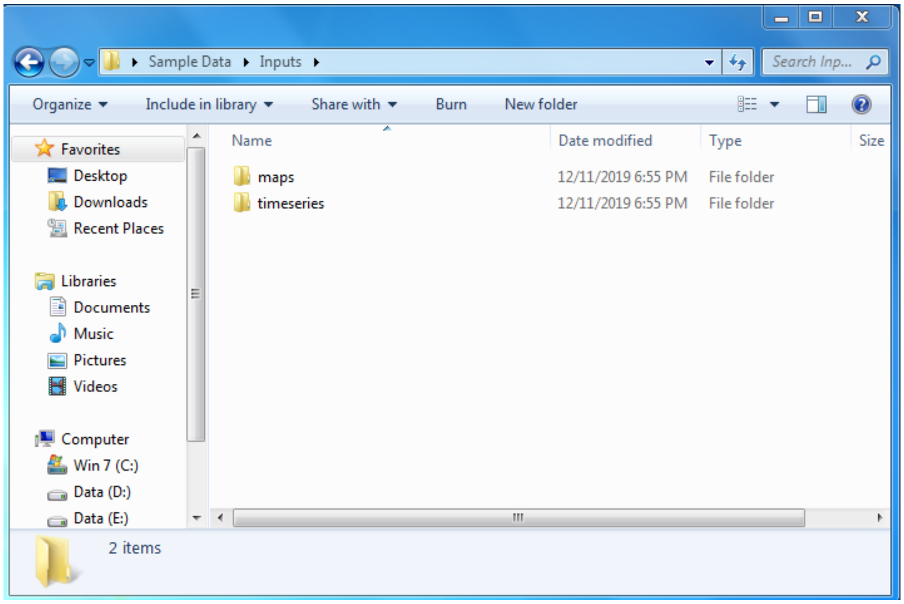


Figure 4. Example of the model working directory structure within the *Inputs* folder used for automatic data loading (autoload)

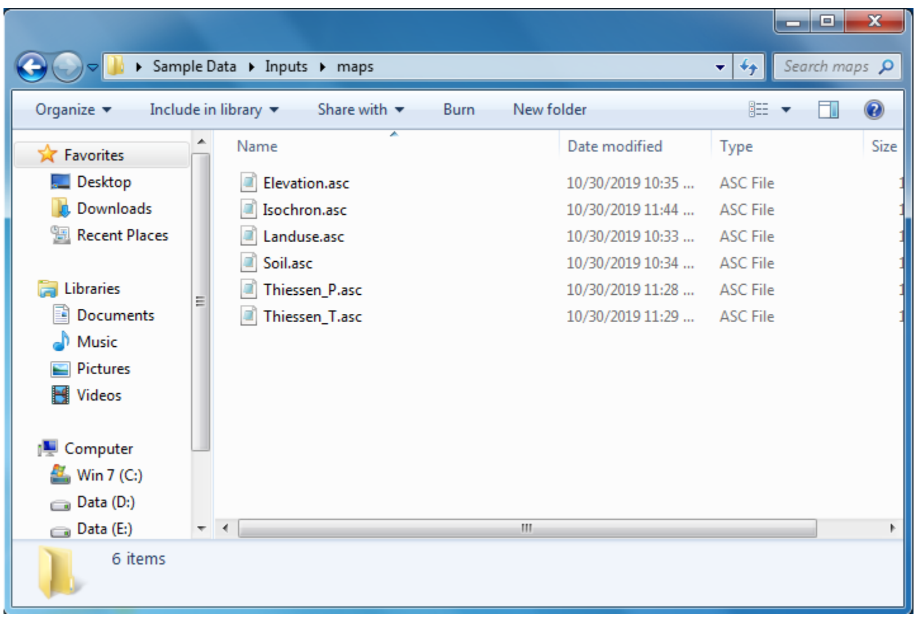


Figure 5. Default filenames for the raster maps in the *maps* subfolder used for automatic data loading (autoload)

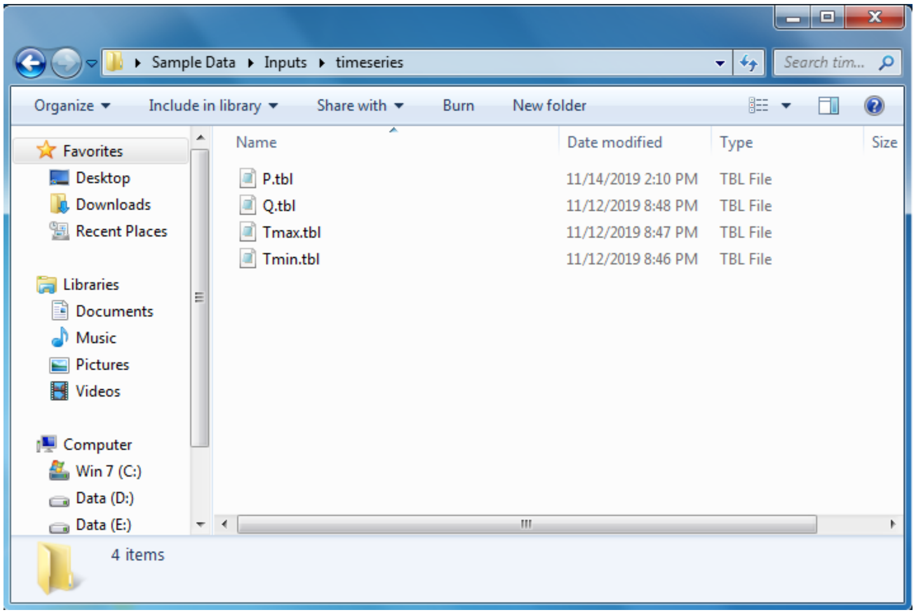


Figure 6. Default time-series filenames recognized by the model for automatic data loading (autoload) in the *timeseries* subfolder

2.1.1. Model Inputs

As mentioned previously, the model inputs consist of raster maps and time-series data. The time-series inputs include **precipitation**, **observed discharge**, **minimum temperature**, and **maximum temperature**. In all time-series files, the first three columns represent the **year**, **month**, and **day**, respectively.

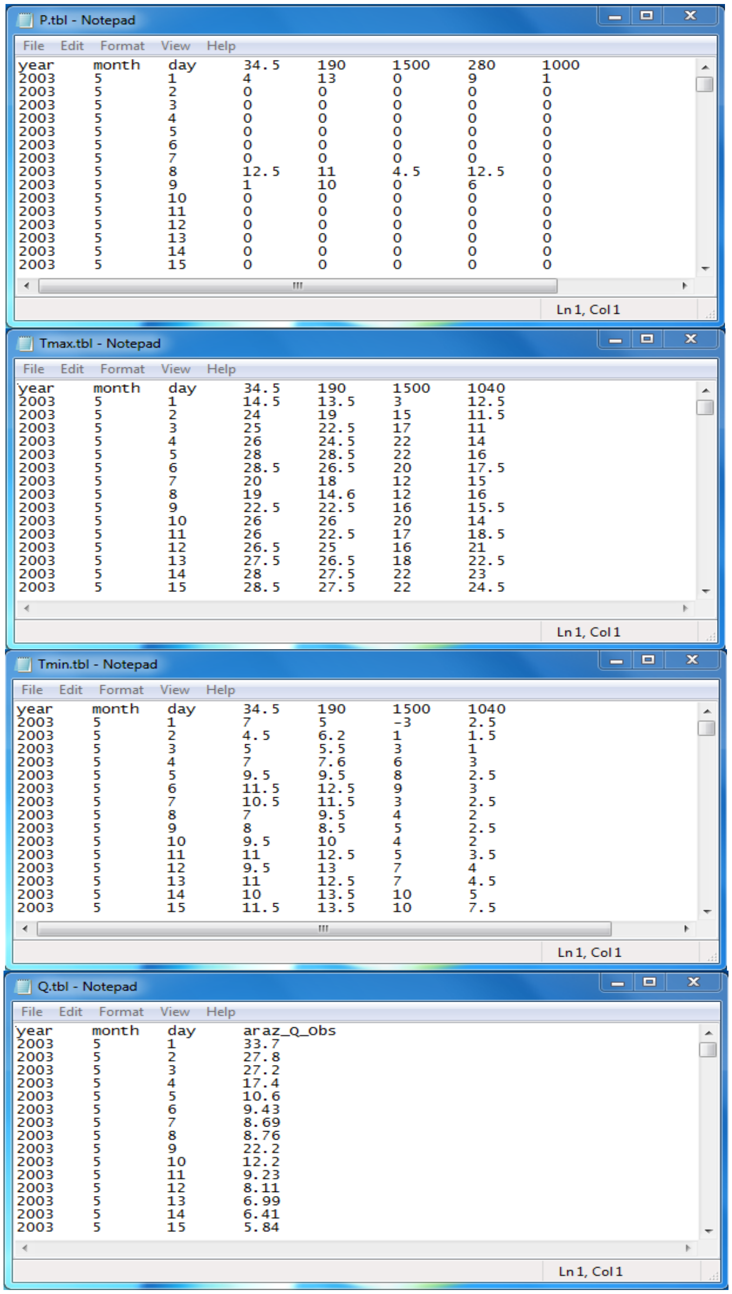
For the discharge file, the fourth column contains the observed discharge values recorded at the gauging station located at the basin outlet. In the precipitation, minimum temperature, and maximum temperature files, the subsequent columns contain the data corresponding to the number of stations used for each variable.

It should be noted that the order of stations in each column must correspond to the station codes defined in the **Thiessen precipitation and temperature maps**. The elevation of each station is provided in the first row (header) corresponding to the data of each station.

Figure 7 shows an example of the model time-series input files.

The raster map inputs include the **Digital Elevation Model (DEM)**, **land-use map**, **soil map**, **isochrone map**, and **Thiessen maps** corresponding to precipitation and temperature stations. The land-use and soil maps are coded according to the reference codes provided in the lookup tables (**Tables 1 and 2**) available in the appendix section.

A
B
C
D



year	month	day	34.5	190	1500	280	1000
2003	5	1	4	13	0	9	1
2003	5	2	0	0	0	0	0
2003	5	3	0	0	0	0	0
2003	5	4	0	0	0	0	0
2003	5	5	0	0	0	0	0
2003	5	6	0	0	0	0	0
2003	5	7	0	0	0	0	0
2003	5	8	12.5	11	4.5	12.5	0
2003	5	9	1	10	0	6	0
2003	5	10	0	0	0	0	0
2003	5	11	0	0	0	0	0
2003	5	12	0	0	0	0	0
2003	5	13	0	0	0	0	0
2003	5	14	0	0	0	0	0
2003	5	15	0	0	0	0	0

year	month	day	34.5	190	1500	1040
2003	5	1	14.5	13.5	3	12.5
2003	5	2	24	19	15	11.5
2003	5	3	25	22.5	17	11
2003	5	4	26	24.5	22	14
2003	5	5	28	28.5	22	16
2003	5	6	28.5	26.5	20	17.5
2003	5	7	20	18	12	15
2003	5	8	19	14.6	12	16
2003	5	9	22.5	22.5	16	15.5
2003	5	10	26	26	20	14
2003	5	11	26	22.5	17	18.5
2003	5	12	26.5	25	16	21
2003	5	13	27.5	26.5	18	22.5
2003	5	14	28	27.5	22	23
2003	5	15	28.5	27.5	22	24.5

year	month	day	34.5	190	1500	1040
2003	5	1	7	5	-3	2.5
2003	5	2	4.5	6.2	1	1.5
2003	5	3	5	5.5	3	1
2003	5	4	7	7.6	6	3
2003	5	5	9.5	9.5	8	2.5
2003	5	6	11.5	12.5	9	3
2003	5	7	10.5	11.5	3	2.5
2003	5	8	7	9.5	4	2
2003	5	9	8	8.5	5	2.5
2003	5	10	9.5	10	4	2
2003	5	11	11	12.5	5	3.5
2003	5	12	9.5	13	7	4
2003	5	13	11	12.5	7	4.5
2003	5	14	10	13.5	10	5
2003	5	15	11.5	13.5	10	7.5

year	month	day	araz_Q_obs
2003	5	1	33.7
2003	5	2	27.8
2003	5	3	27.2
2003	5	4	17.4
2003	5	5	10.6
2003	5	6	9.43
2003	5	7	8.69
2003	5	8	8.76
2003	5	9	22.2
2003	5	10	12.2
2003	5	11	9.23
2003	5	12	8.11
2003	5	13	6.99
2003	5	14	6.41
2003	5	15	5.84

Figure 7. Examples of the model time-series input data: precipitation (A), maximum temperature (B), minimum temperature (C), and observed discharge (D)

2.1.1.1. Preparation of the Isochrone Map Using GIS (Based on Travel Time and Flow Velocity)

To calculate the basin travel-time map, the flow velocity is first determined in a GIS environment using **Manning's equation** (Equation 1).

$$v_i = \frac{1}{n_i} R_i^{\frac{2}{3}} S_i^{\frac{1}{2}}$$ **Eq.1**

where v_i is the flow velocity (m/s), R_i is the hydraulic radius (m), S_i is the slope (m/m), and n_i is Manning's roughness coefficient ($Sm^{-1/3}$), which depends on the land use and channel characteristics. The hydraulic radius is calculated using the power-law relationship (Equation 2). In this equation, the hydraulic radius is related to the upstream contributing area and represents the approximate behavior of each cell as well as the channel bed geometry.

$$R_i = a_p (A_i)^{b_p}$$ **Eq.2**

where $A_i$is the upstream drainage area of each cell (km^2), a_p is the network constant, and b_p is an exponent related to channel geometry. Both parameters depend on the discharge return period. In this study, the values of a_p and b_p are assumed to be **0.09** and **0.5**, respectively. The upstream drainage area map for each cell is obtained in **ArcGIS** (flowaccumulation*cellsize^2).

After calculating the flow velocity using Manning's equation, the wave celerity is determined using Equation 3.

$$c_i = \frac{5 v_i}{3}$$ **Eq.3**

Finally, after calculating the wave celerity, the travel time is determined using Equation 4.

$$T_t = \frac{\mathrm{L}}{c_i}$$ **Eq.4**

where c_i is the wave celerity (m/s), and L is the flow length (m) obtained from the **Flow Length** map. After calculating the travel time based on wave celerity, the **isochrones (lines of equal travel time)** are generated from the travel-time map.

2.2. Second Model Tab (Parameters/Simulation)

The second model tab (**Parameters/Simulation**) includes the model parameters and the **Run** operation, or alternatively, the model settings and simulation configuration (Figure 8). In this tab, users define the model parameters; in fact, the selection of optimal parameter values and model calibration are performed in this section.

The allowable ranges of the model parameters are provided in **Table 3** in the Appendix section.

To improve the efficiency of the model calibration process, this tab includes a **check box** named **Preprocessing data**. This option is activated only during the first model run and enables the preprocessing of all input data and raster maps. In subsequent runs, the option is automatically disabled, allowing model calibration to be performed at a higher speed.

It should be noted that the model has a very high computational efficiency. Including the preprocessing step, the model can be executed on a daily time step for a 10-year simulation period in less than one minute.

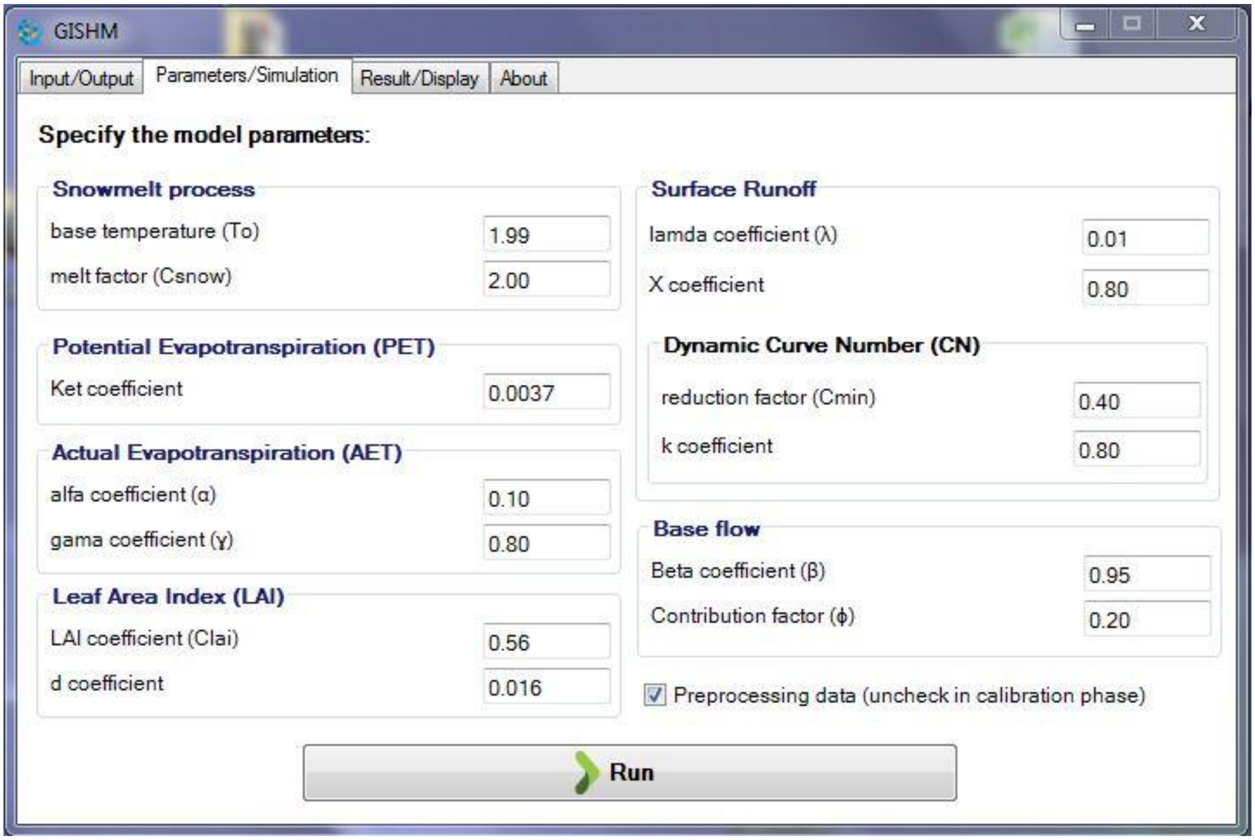


Figure 8. The second model tab (Parameters/Simulation), showing the model parameters and simulation operation

Considering the implementation of error handling for potential issues during data input and model execution, if users unintentionally make errors while entering data or defining model parameters, an error message will be displayed in this tab during the model execution stage. This message informs users about the corresponding error in each section.

If no errors are detected, or if all identified errors have been resolved, the model starts running and performs the simulation by activating the **Run** operation. During this stage, the model progress bar (**ProgressBar**) appears in this tab and continuously reports the simulation progress until completion (**Figure 9**).

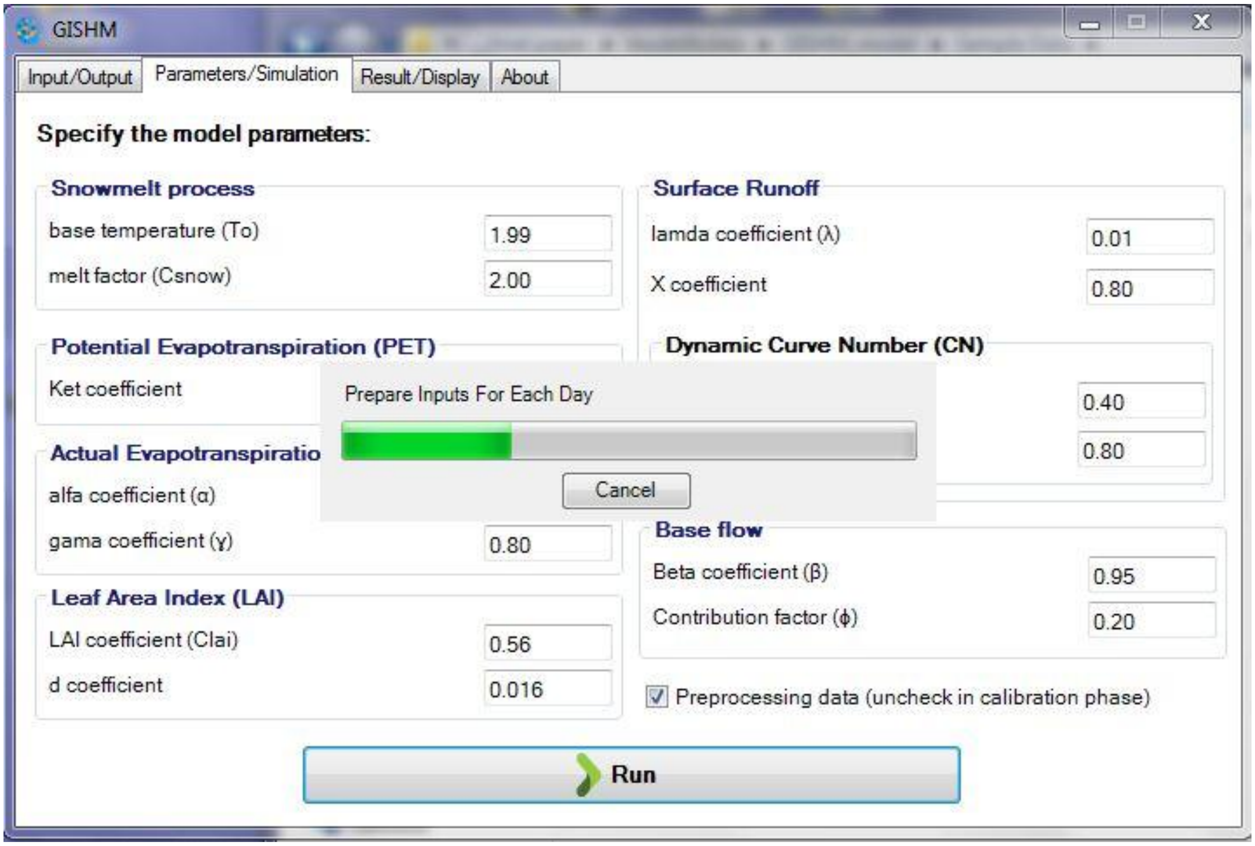


Figure 9. Model progress bar

2.3. Third Model Tab (Result/Display)

The third model tab (**Result/Display**) includes the model performance evaluation criteria results and the display of the runoff hydrograph at the basin outlet (Figure 10). The performance criteria used in the model include the **Nash–Sutcliffe Efficiency (NSE)**, **Bias (model deviation)**, **Kling–Gupta Efficiency (KGE)**, **Modified Kling–Gupta Efficiency (MKGE)**, **Coefficient of Determination (R^2)**, and **Weighted Correlation Coefficient**. The results of all these performance metrics are displayed in this tab.

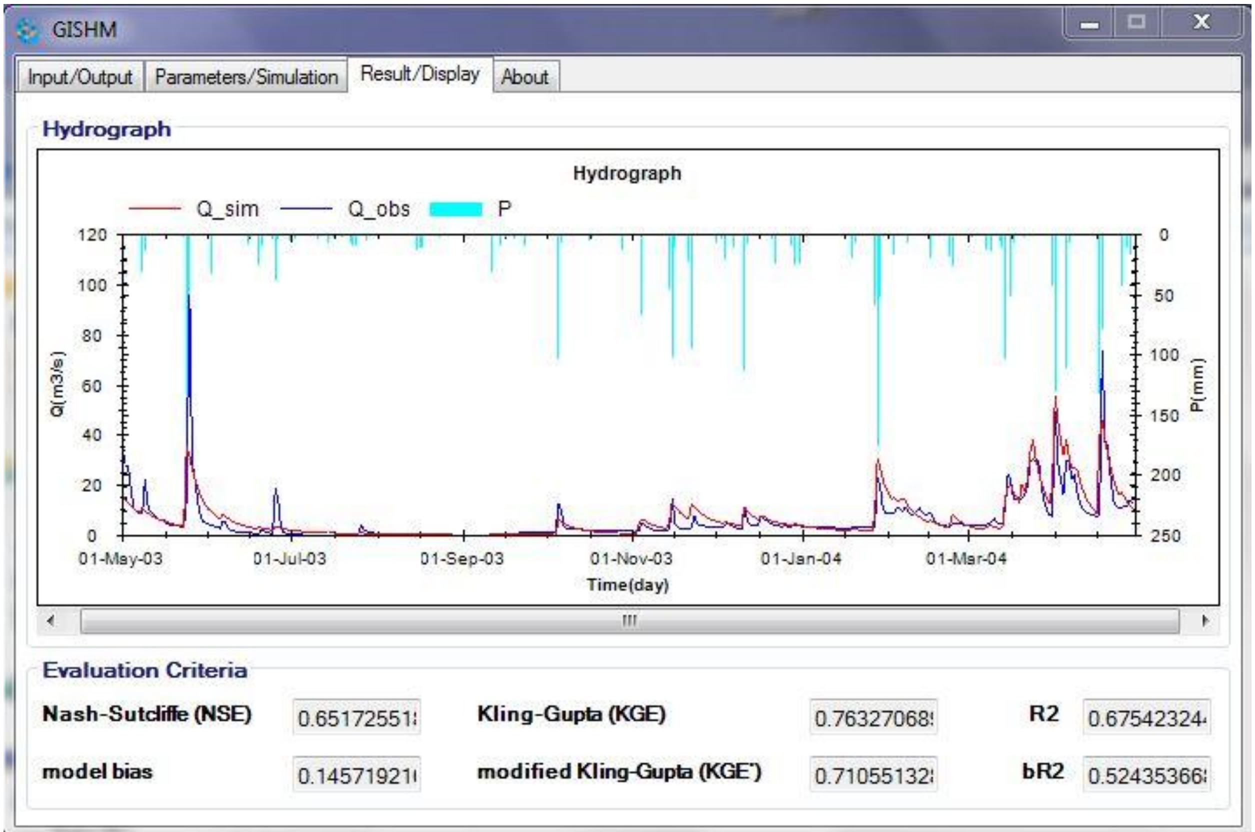


Figure 10. The third model tab (Result/Display): Model performance criteria results and the outlet runoff hydrograph.

2.4. Fourth Model Tab (About)

The fourth and final model tab (**About**) provides general information about the model and the research team (Figure 11).

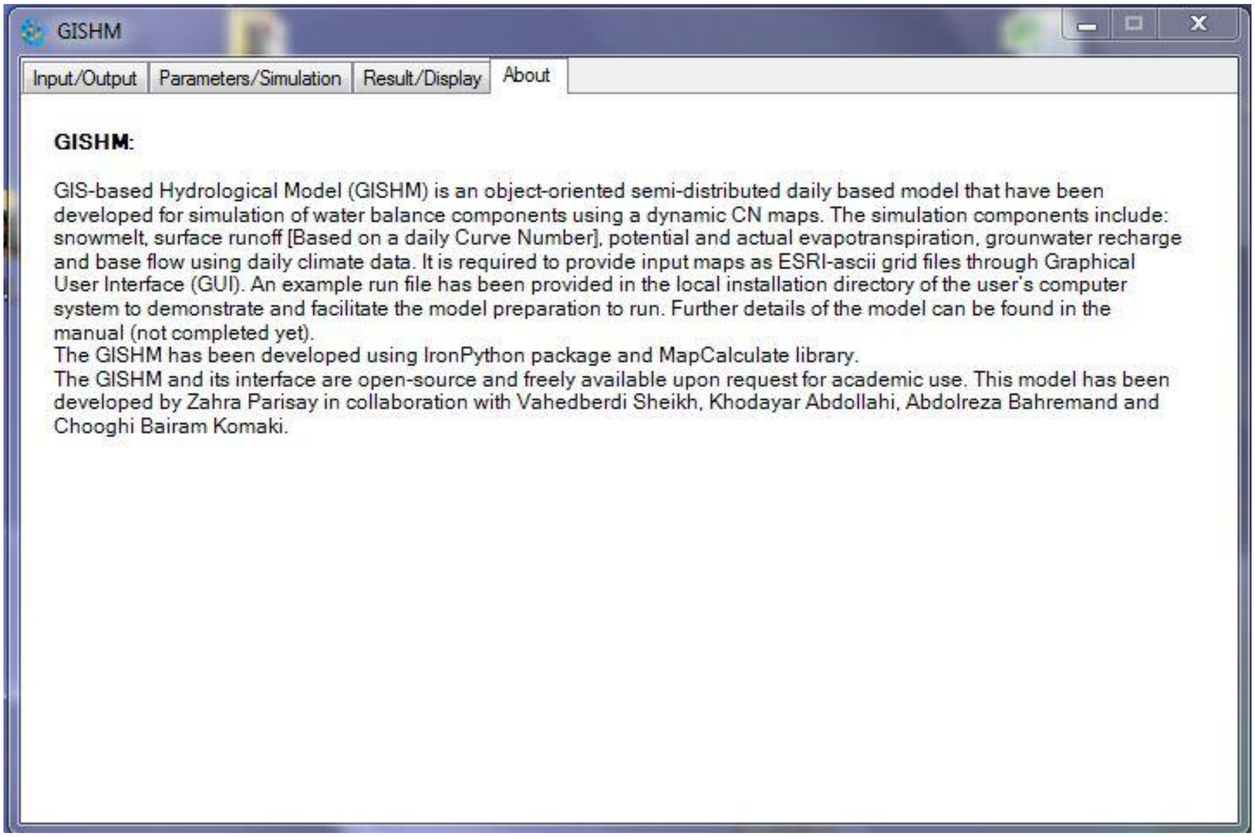


Figure 11. The fourth model tab (About), showing general information about the model and the research team

Appendix

1. Model Lookup Tables

To generate the basin **curve number (CN) maps**, the **average leaf area index (LAI)**, the **maximum basin leaf area index**, and the **vegetation cover fraction** based on soil and land-use maps, standard lookup tables have been defined. Users can modify and adjust the values provided in these tables according to the characteristics of the study area (**Tables 1 and 2**).

Table 1. Model lookup table based on soil classification codes.

Code	Soil	Hydrologic Soil Group (HSG)
1	sand	A
2	lomy sand	A
3	sandy loam	B
4	loam	B
5	silt loam	C
6	sandy clay loam	C
7	clay loam	C
8	silty clay loam	D
9	sandy clay	D
10	silty clay	D
11	clay	D

Table 2. Model lookup table based on land-use classification codes.

Code	Landuse	Curve number (Cronshey, 1986)				Cp	LAI_{avg}	LAI_{max}
		D	C	B	A			
1	Streets and roads	98	98	98	98	0.4	0.30	0.78
2	Industrial	93	91	88	81	0.3	0.30	0.90
3	Residential area (65% impervious area)	92	90	85	77	0.2	0.34	1.00
4	Residential area (20% impervious area)	84	79	68	51	0.3	0.39	1.50
5	Bare Lands	94	91	86	77	0.05	0.15	0.50
6	Irrigated Agriculture	85	82	74	59	0.8	0.70	2.50
7	Rainfed Agriculture	92	88	83	73	0.7	0.60	2.40
8	Pasture, grassland or rangeland (Poor)	89	68	79	68	0.5	0.50	1.40
9	Pasture, grassland or rangeland (Fair)	84	79	69	49	0.6	0.75	1.80
10	Pasture, grassland or rangeland (Good)	80	74	61	39	0.7	0.95	2.00
11	Wood-grass combination (**orchard** or tree farm) (Poor)	86	82	73	57	0.7	0.75	2.20
12	Wood-grass combination (**orchard** or tree farm) (Fair)	82	76	65	43	0.8	1.30	2.60
13	Wood-grass combination (**orchard** or tree farm) (Good)	79	72	58	32	0.9	1.50	2.80
14	Forest land (Poor)	83	77	66	45	0.8	1.60	3.30
15	Forest land (Fair)	79	73	60	36	0.9	1.90	3.70
16	Forest land (Good)	77	70	55	30	1	2.00	3.90
17	Salty land	88	85	77	63	0	0.20	0.70
18	Water body	100	100	100	100	0	0	0

2. Model Parameter Ranges

Table 3. Ranges of model parameters

Parameters	**Parameter Range**
Threshold temperature (T_o)	-3 - 2
melt factor (C_{snow})	-3 - 3.2
Potential evapotranspiration coefficient(K_{ET})	0 - 1
Actual evapotranspiration parameter (α)	0 - 1
Actual evapotranspiration parameter (γ)	0 - 2
Daily leaf area index parameter (C_{lai})	0 - 1
Daily leaf area index parameter (d)	0 - 1
Initial abstraction ratio (λ)	0 - 1
Delay factor (X)	0 - 1
Daily curve number reduction parameter (C_{min})	0 - 1
Daily curve number parameter (k)	0 - 1
Base-flow storage parameter (β)	0 - 1
Recharge contribution parameter to current base-flow ($\emptyset$)	0 - 1